

KO LANTA, Thailand

WMO: 485660

Lat: 7.533N

Lon: 99.050E

Elev: 10

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	74.3	75.2	66.8	98.6	83.6	68.2	103.8	83.2	14.5	83.4	12.9	83.3	2.0	90	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.4	93.1	79.8	92.0	80.0	91.0	80.0	84.0	89.2	83.4	88.6	82.7	88.0	5.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.6	170.1	87.8	81.9	165.9	87.2	81.3	162.5	86.6	48.2	89.5	47.4	88.6	46.6	88.2	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.3	18.7	16.3	71.8	95.2	1.6	1.6	70.6	96.4	69.7	97.3	68.9	98.3	67.7	99.5		
(5)			69.5	84.4	1.4	1.6	68.5	85.6	67.7	86.5	66.9	87.5	65.9	88.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.1	82.6	83.7	84.4	84.6	84.1	83.4	82.9	83.0	82.3	82.0	82.1	82.1
	DBStd	2.00	1.81	1.66	2.00	1.87	1.79	1.76	2.07	1.88	1.54	1.52	1.77	1.77
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12079	1010	945	1067	1037	1056	1001	1021	1022	969	993	963	995
	CDD65	6604	545	525	602	587	591	551	556	557	519	528	513	530
	CDH74	76452	5825	6002	7297	7305	7487	6777	6638	6698	5958	5653	5291	5521
CDH80	28880	2002	2404	3183	3227	3197	2717	2553	2520	1990	1747	1607	1733	
Wind	WSAvg	5.2	3.8	4.2	3.6	3.3	5.7	6.7	7.5	8.8	7.1	5.2	2.5	3.9
Precipitation	PrecAvg	86.2	1.2	1.1	3.5	5.0	8.5	10.0	11.0	12.6	12.2	11.5	6.5	3.2
	PrecMax	105.4	6.1	3.3	12.8	10.7	18.5	16.9	18.6	21.9	18.1	20.0	12.1	8.9
	PrecMin	68.5	0.0	0.0	0.0	0.8	3.3	4.7	5.0	2.2	6.3	3.5	2.6	0.3
	PrecStd	9.9	1.5	1.1	3.6	2.9	3.3	3.2	3.1	4.9	3.3	3.9	2.7	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.8	93.6	94.9	94.4	93.2	91.2	90.5	89.9	89.8	90.0	91.0	91.2
		MCWB	78.3	77.1	78.9	81.3	82.6	82.6	81.7	81.5	81.7	80.7	80.5	78.5
	2%	DB	90.5	92.3	93.1	92.7	91.4	89.8	89.0	88.6	88.2	88.5	89.5	89.7
		MCWB	78.0	77.4	79.3	81.4	81.9	81.8	81.0	80.9	80.8	80.4	80.1	78.6
	5%	DB	89.3	91.0	91.7	91.5	90.1	88.8	88.0	87.7	87.3	87.3	88.3	88.5
		MCWB	77.8	77.7	79.7	81.2	81.7	81.3	80.6	80.5	80.3	80.0	79.8	78.4
	10%	DB	87.9	89.6	90.4	90.3	88.9	87.8	87.1	86.9	86.3	86.2	87.1	87.2
		MCWB	77.7	77.7	79.6	80.8	81.4	80.8	80.2	80.1	80.0	79.5	79.3	78.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.6	82.4	83.7	84.5	84.8	84.6	83.6	83.5	83.4	82.7	82.7	82.1
		MCDB	87.5	89.6	90.2	90.5	89.6	88.9	88.2	87.7	88.2	87.1	88.5	87.9
	2%	WB	81.6	81.0	82.6	83.5	84.0	83.3	82.6	82.5	82.2	81.8	81.6	80.8
		MCDB	87.2	88.5	89.6	89.9	89.1	88.0	87.4	86.6	86.6	86.3	87.4	86.9
	5%	WB	80.2	79.9	81.7	82.7	83.2	82.4	81.8	81.8	81.4	81.0	80.7	79.9
		MCDB	86.0	87.7	88.9	89.3	88.1	87.4	86.7	86.1	85.9	85.6	86.5	85.8
	10%	WB	79.0	78.8	80.7	81.7	82.2	81.6	81.0	81.2	80.7	80.4	79.9	79.0
		MCDB	85.0	86.9	88.2	88.4	87.1	86.6	86.1	85.7	85.1	85.0	85.4	84.8
Mean Daily Temperature Range	5% DB	MDBR	11.4	12.9	12.6	11.4	9.0	8.5	8.1	7.2	7.8	8.6	9.8	10.5
		MCDBR	12.9	14.0	14.1	13.0	11.2	10.2	9.1	8.5	9.7	10.6	11.8	12.2
		MCWBR	4.7	4.7	5.1	4.9	4.7	4.8	4.3	4.0	4.6	4.8	4.6	4.6
	5% WB	MCDBR	11.9	13.0	12.9	12.1	10.2	9.8	8.6	7.7	9.0	9.7	11.2	11.2
		MCWBR	5.9	6.0	6.0	5.6	5.7	5.3	4.9	4.4	5.0	5.0	5.0	5.4
Clear-Sky Solar Irradiance	taub		0.461	0.475	0.486	0.474	0.451	0.456	0.459	0.453	0.441	0.444	0.450	0.457
	taud		2.314	2.245	2.253	2.293	2.391	2.399	2.377	2.383	2.411	2.404	2.372	2.341
	Ebn at Noon		272	271	268	267	268	263	263	269	276	276	272	270
	Edh at Noon		42	46	46	43	38	37	38	39	38	38	39	40
All-Sky Solar Radiation	RadAvg		1667	1946	1956	1856	1624	1567	1552	1587	1539	1499	1437	1438
	RadStd		119	110	184	115	106	105	86	73	56	115	98	135

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		+0.34	+0.45	N/A	N/A	+0.34	+0.38	N/A	N/A	+133	+135

Nomenclature: See separate page